

# Mean Field Model for Protein Crowding

Bridging the Scales Inside the Cell – Görlitz 2026

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# The Model's Code

The implementation of the mean field model is available on GitHub:

<https://github.com/gillolgenblum/CrowdingModeWorkshop/tree/main>



# Code Repository Main Page

## Installation

You can install this package in editable mode using `pip`:

```
pip install -e .
```

To install with development dependencies (e.g., Jupyter for running notebooks):

```
pip install -e .[dev]
```

To install with app dependencies (e.g., Streamlit for the local web app):

```
pip install -e .[app]
```

## References

1. Sapir, L.; Harries, D. Macromolecular Stabilization by Excluded Cosolutes: Mean Field Theory of Crowded Solutions. *Journal of Chemical Theory and Computation* **2015**, *11* (7), 3478-3490.
2. Sapir, L.; Harries, D. Macromolecular Compaction by Mixed Solutions: Bridging versus Depletion Attraction. *Current Opinion in Colloid & Interface Science* **2016**, *22*, 80-87.
3. Stewart, C. J.; Olgenblum, G. I.; Propst, A.; Harries, D.; Pielak, G. J. Resolving the Enthalpy of Protein Stabilization by Macromolecular Crowding. *Protein Science* **2023**, *32* (3), e4573.
4. Olgenblum, G. I.; Carmon, N.; Harries, D. Not Always Sticky: Specificity of Protein Stabilization by Sugars Is Conferred by Protein-Water Hydrogen Bonds. *Journal of the American Chemical Society* **2023**, *145* (42), 23308-23320.
5. Olgenblum, G. I.; Stewart, C. J.; Redvanly, T. W.; Young, O. M.; Lauzier, F.; Hazlett, S.; et al.; Harries, D. Crowding beyond Excluded Volume: A Tale of Two Dimers. *Protein Science* **2025**, *34* (4), e70062.
6. Redvanly, T. W.; Olgenblum, G. I.; Young, O. M.; Goldenberg, Y.; Stewart, C. J.; Harries, D.; Pielak, G. J. Sugar-Protein Interactions Control Protein-Complex Stability in Crowded Ficoll and Dextran Solutions. *Protein Science* **2026**, *35* (1), e70416.

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# Code Repository Content

 CrowdingModeWorkshop Public Pin 👁

main 1 Branch 0 Tags Go to file T + <> Code

 gillolgenblum new exercise and added question marks 37ee219 · 3 hours ago 22 Commits

|                                                                                                      |                                                              |             |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------|
|  app                | Replace Greek Unicode with LaTeX in E1/E2 notebooks; fix ... | 5 days ago  |
|  src/crowding       | Corrected notebooks                                          | last week   |
|  tutorials          | new exercise and added question marks                        | 3 hours ago |
|  .gitignore         | Edits to solution notebooks                                  | 5 days ago  |
|  LICENSE           | chore: add standard Python .gitignore and GPLv3 license      | last week   |
|  README.md        | Replace Greek Unicode with LaTeX in E1/E2 notebooks; fix ... | 5 days ago  |
|  output.png       | Replace Greek Unicode with LaTeX in E1/E2 notebooks; fix ... | 5 days ago  |
|  pyproject.toml   | Initial commit for Workshop Binary App                       | last week   |
|  requirements.txt | Initial commit for Workshop Binary App                       | last week   |

Streamlit  
app

Scientific  
code

Tutorials  
and  
exercises

# Streamlit App

<https://crowdingmodeworkshop-jg6xuovuytncequyqqdhqh.streamlit.app/>



**Zzzz**

This app has gone to sleep due to inactivity. Would you like to wake it back up?

Yes, get this app back up!

---

If you believe this is a bug, please [contact us](#) or [visit the Streamlit forums](#).

# Streamlit App

### Model Parameters

---

>  Save/Load Session

---

### Protein

SASA

419.0 - +

### Temperature

☒ Include Temperature

Temperature (K)

298.15 - +



## FH Crowding Thermodynamic Model

Flory-Huggins model for protein-cosolute crowding thermodynamics. Choose a mode below to simulate or fit experimental data.

 **Mode A — Plug-and-Play Simulation**  
1. Choose model (sidebar) → 2. Set parameters → 3. Run simulation → 4. Plot & explore → 5. Download results

 **Mode B — Fit to Experimental Data**  
1. Choose model → 2. Upload data (below) → 3. Set initial parameters → 4. Run fit → 5. Plot & review diagnostics → 6. Download

 **Quick Tips:**

- Click **Apply Parameters** (in the left sidebar) whenever you change model parameters.
- Fit from Scratch** when changing parameters or uploading new data files to ensure accurate results.

 **Experimental Data & Fitting Hub**

< Manage app

# Scientific Code

 `__init__.py`

---

 `binary.py`

---

Main workflows

 `constants.py`

---

Constants and unit conversions

 `cosolute.py`

---

Cosolute object and relevant functions

 `plotting.py`

---

Preconfigured visualization tools

 `protein.py`

---

Protein object

 `statistics.py`

---

Cross validating function for error estimations

# Tutorials and Exercises

## Core Tutorials and Exercises

1. `00_getting_started.ipynb` : Installation, scientific background, and setup.
2. `01_plug_and_play_simulation.ipynb` : Running forward binary simulations.
  - `E1_forward_simulation_exercise.ipynb` : Forward simulation exercises.
3. `02_fit_experimental_data.ipynb` : Fitting soft interaction parameters to binary data.
  - `E2_fitting_exercise.ipynb` : Data fitting exercises.
4. `03_parameter_sensitivity_and_interpretation.ipynb` : Understanding the physical effects of SASA and Flory-Huggins  $\chi$ .
5. `04_exporting_results_and_figures.ipynb` : Extracting datasets and tweaking matplotlib charts for publication.
6. `05_polymer_scaling_and_domain_size.ipynb` : Advanced simulation of polymer scaling behavior.
7. `06_statistical_analysis_and_uncertainty.ipynb` : Monte-Carlo Cross-Validation for uncertainty analysis.
  - `E3_uncertainty_analysis_exercise.ipynb` : Uncertainty analysis exercises.
  - `E4_determination_of_parameters_exercise.ipynb` : Determination of FH parameters from experimental data.

- All tutorials and exercises are written in Jupyter notebooks and can be run via Google Colab.
- You can work on your own pace.
- Every 5-10 min we will go through a tutorial notebook together.
- Every ~20 min we will go through a solution to an exercise together.



# Tutorials and Exercises

To work on colab instead of  
locally/hpc press



## Getting Started with CrowdingModeWorkshop

Welcome to the `CrowdingModeWorkshop` tutorials! The `crowding` package implements the Flory-Huggins based thermodynamic model for protein stability in crowded environments.

# Tutorials and Exercises

Some exercises require you  
to fill the ???

```
# Initialize the binary model
model = cr.CrowdingModel(
    protein=???,
    cosolute=???,
    eps=???,          # Soft interactions parameter
    epsTS=???,        # Entropic component of soft interactions
    phiC_max=???,     # Maximum volume fraction to simulate
    dphiC=???,        # Step size for the grid
    T=298.15          # Temperature in Kelvin
)
```

Others require you to write  
the entire code by yourself

```
# YOUR CODE HERE
```

# Enjoy!

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  - `E4_determination_of_parameters_exercise.ipynb` : Determination of FH parameters from experimental data.